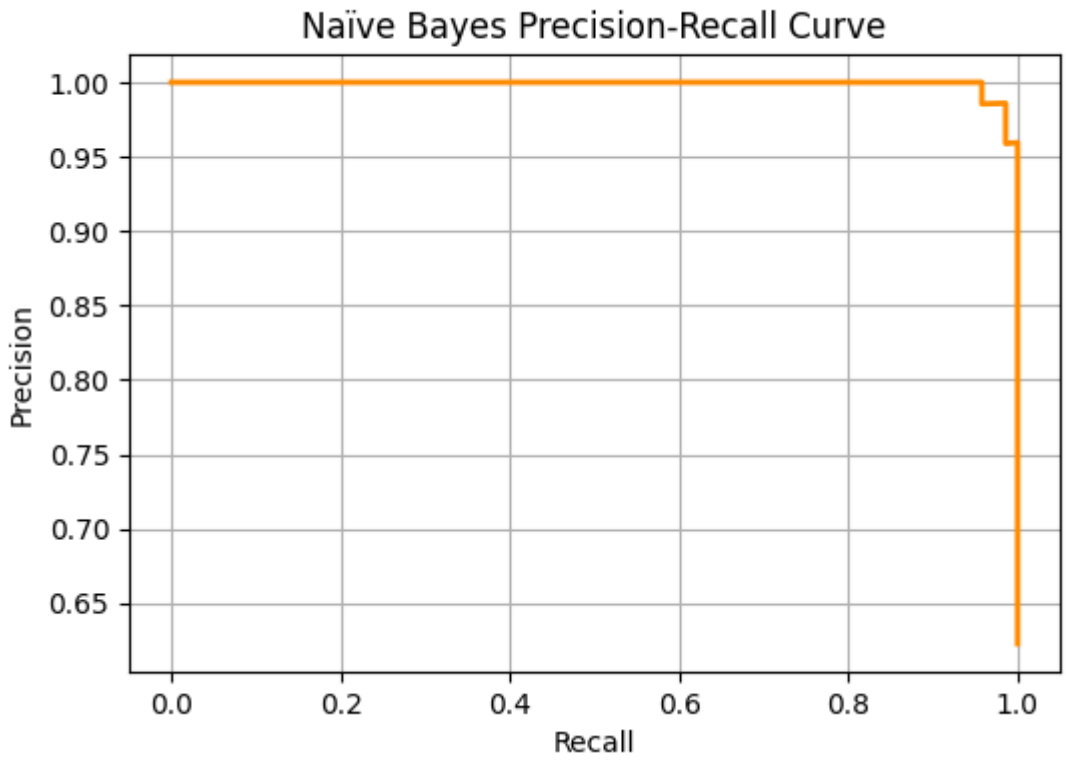
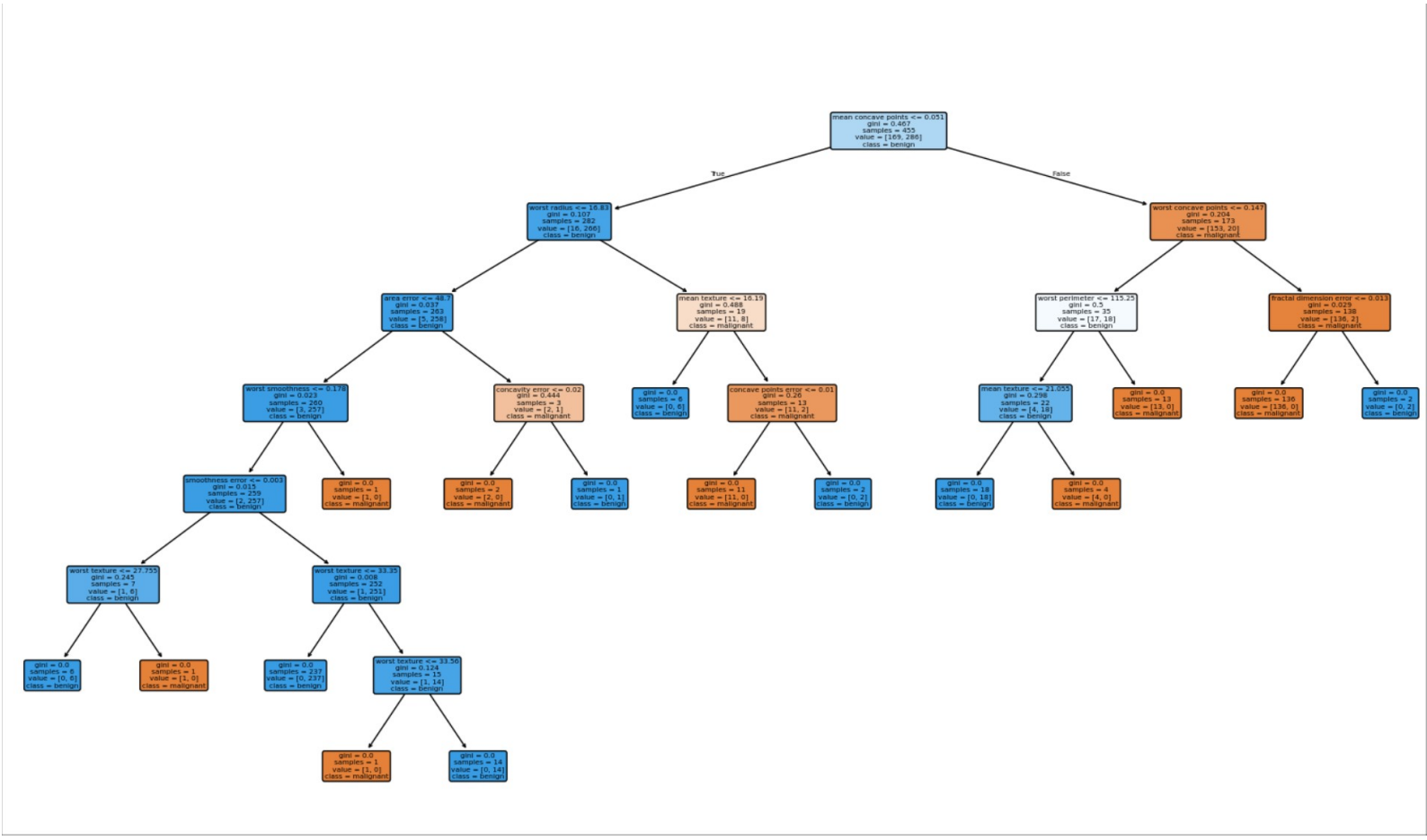
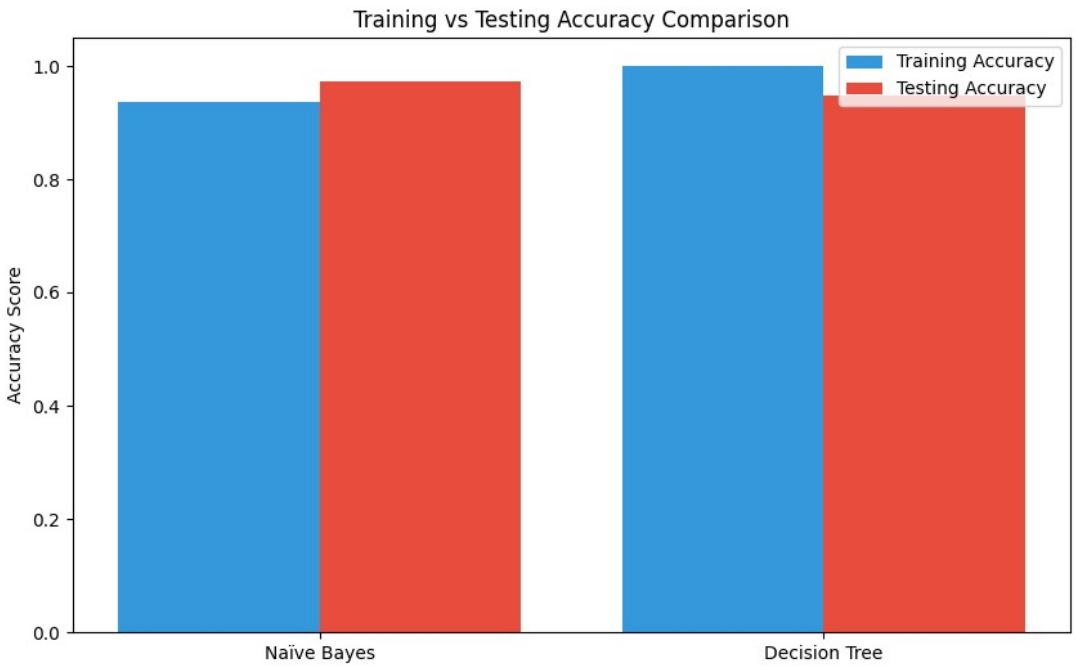
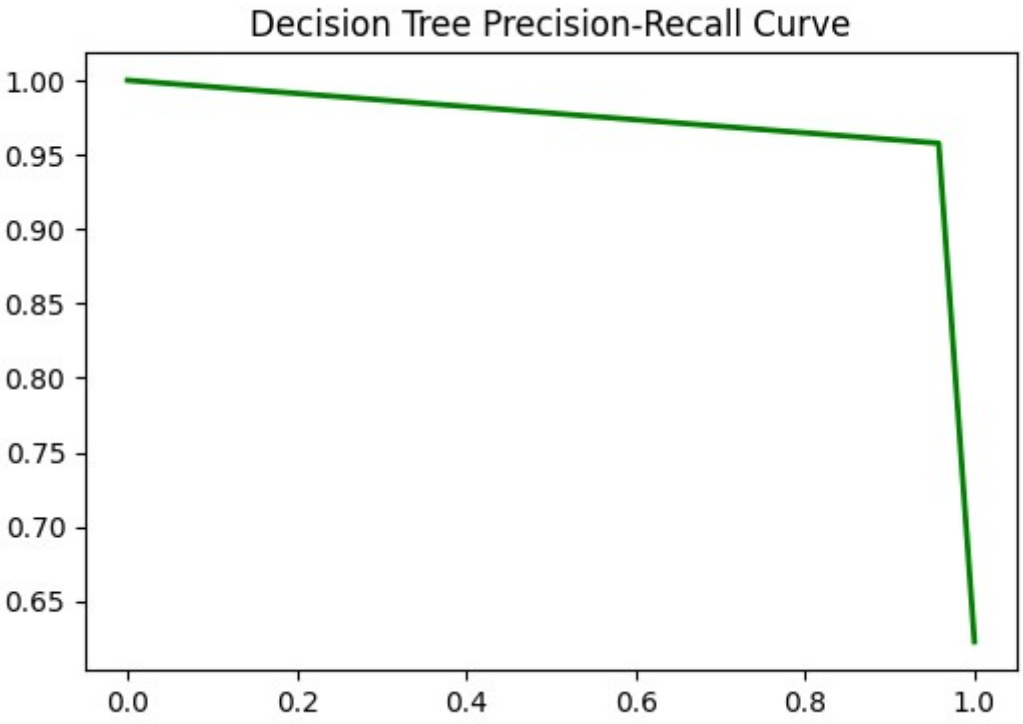
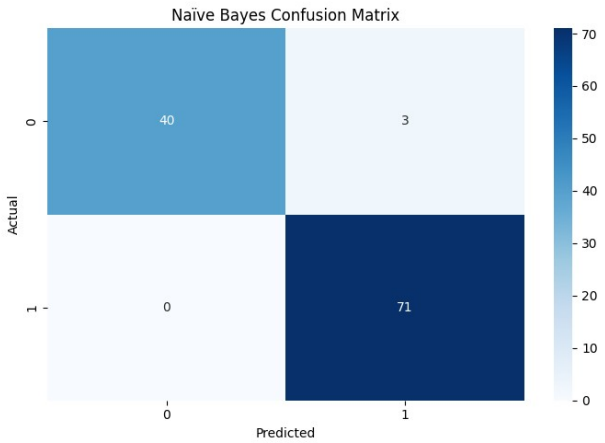

Naïve Bayes Precision-Recall Curve
Precision
Recall
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.0
0.2
0.4
0.6
0.8
1.0

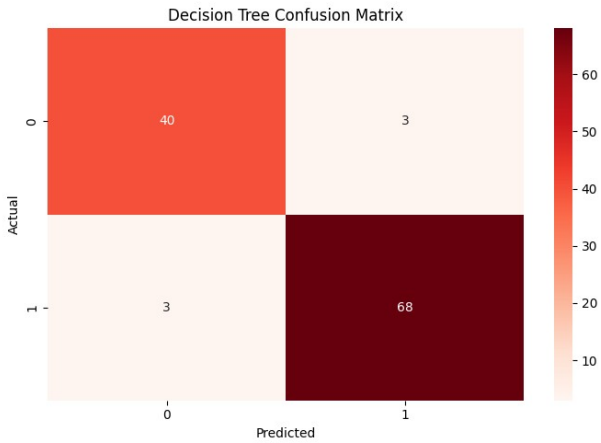

mean concave points <= 0.051
gini = 0.467
samples = 455
value = [169, 286]
class = benign
True
False
worst radius <= 16.83
gini = 0.107
samples = 282
value = [16, 266]
class = benign
worst concave points <= 0.147
gini = 0.204
samples = 173
value = [153, 20]
class = malignant
area error <= 48.7
gini = 0.037
samples = 263
value = [5, 258]
class = benign
mean texture <= 16.19
gini = 0.488
samples = 19
value = [11, 8]
class = malignant
worst perimeter <= 115.25
gini = 0.5
samples = 35
value = [17, 18]
class = benign
fractal dimension error <= 0.013
gini = 0.029
samples = 138
value = [136, 2]
class = malignant
worst smoothness <= 0.178
gini = 0.023
samples = 260
value = [3, 257]
class = benign
concavity error <= 0.02
gini = 0.444
samples = 3
value = [2, 1]
class = malignant
gini = 0.0
samples = 6
value = [0, 6]
class = benign
concave points error <= 0.01
gini = 0.26
samples = 13
value = [11, 2]
class = malignant
mean texture <= 21.055
gini = 0.298
samples = 22
value = [4, 18]
class = benign
gini = 0.0
samples = 13
value = [13, 0]
class = malignant
gini = 0.0
samples = 136
value = [136, 0]
class = malignant
gini = 0.0
samples = 2
value = [0, 2]
class = benign
smoothness error <= 0.003
gini = 0.015
samples = 259
value = [2, 257]
class = benign
gini = 0.0
samples = 1
value = [1, 0]
class = malignant
gini = 0.0
samples = 2
value = [2, 0]
class = malignant
gini = 0.0
samples = 1
value = [0, 1]
class = benign
gini = 0.0
samples = 11
value = [11, 0]
class = malignant
gini = 0.0
samples = 2
value = [0, 2]
class = benign
gini = 0.0
samples = 18
value = [0, 18]
class = benign
gini = 0.0
samples = 4
value = [4, 0]
class = malignant
worst texture <= 27.735
gini = 0.245
samples = 7
value = [1, 6]
class = benign
worst texture <= 33.25
gini = 0.008
samples = 252
value = [1, 251]
class = benign
gini = 0.0
samples = 6
value = [0, 6]
class = benign
gini = 0.0
samples = 1
value = [1, 0]
class = malignant
gini = 0.0
samples = 237
value = [0, 237]
class = benign
worst texture <= 33.56
gini = 0.124
samples = 15
value = [1, 14]
class = benign
gini = 0.0
samples = 1
value = [1, 0]
class = malignant
gini = 0.0
samples = 14
value = [0, 14]
class = benign




```
❯ python l5_e1.py; python3 l5_e2.py; python l5_e3.py
--- Experiment 1: Naïve Bayes Results ---
Accuracy: 0.9737

Confusion Matrix:
 [[40  3]
 [ 0 71]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.93      0.96        43
           1       0.96      1.00      0.98        71

    accuracy                           0.97       114
   macro avg       0.98      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114

/home/abhinav/kaam/MLLab/l5_e1.py:34: UserWarning: FigureCanvasAgg is non-interactive, and
thus cannot be shown
  plt.show()
--- Experiment 2: Decision Tree Results ---
Accuracy: 0.9474

Confusion Matrix:
 [[40  3]
 [ 3 68]]
Tree visualization saved as ex2_decision_tree.png
PR Curve saved as ex2_pr_curve.png
Saved: ex3_accuracy_comparison.png
Saved: ex3_roc_comparison.png
Saved: ex3_cm_heatmaps.png
```


Decision Tree Precision-Recall Curve
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.0
0.2
0.4
0.6
0.8
1.0


Naïve Bayes Confusion Matrix
Actual
Predicted
0
1
40
3
0
71
70
60
50
40
30
20
10
0


Decision Tree Confusion Matrix
Actual
Predicted
0
1
40
3
3
68
60
50
40
30
20
10